TimberGrade

AI-Based Wooden Plank Defect Detection and Quality Grading System

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Abstract

Quality inspection of sawn timber is an essential step in the wood processing and timber trade industry, as surface defects such as knots, cracks, rot and stains directly determine the grade, usability and market value of a wooden plank. At present, this inspection is largely performed manually by trained workers, which makes the process slow, inconsistent and prone to human error, especially when large volumes of timber must be checked. Disagreements between buyers and sellers over the quality grade of a shipment are also common.

This project presents **TimberGrade**, an AI-based system that automatically detects surface defects on wooden planks and assigns a quality grade. The system uses **YOLOv8**, a modern and widely used deep learning model for object detection, trained on a publicly available dataset of labelled wood surface images to locate and classify defects such as knots, cracks, rot and stains. Each detected defect is highlighted with a bounding box and a confidence score. A rule-based grading module then evaluates the number, type and size of the defects and classifies the plank as **Grade A, B or C**, along with its recommended use, such as furniture, construction or packaging.

The application is built entirely with open-source tools, including **Python, OpenCV, Ultralytics YOLOv8, Streamlit and SQLite**. It gives users a simple web interface for uploading plank images, viewing detection results, and tracking inspection history through a dashboard. TimberGrade aims to make timber quality inspection faster, more consistent and more objective, reducing manual effort and supporting fair trade in the timber industry.

Keywords: Wood Defect Detection, Timber Grading, YOLOv8, Object Detection, Deep Learning, Computer Vision, OpenCV, Streamlit